

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO4 Assessment Tool 1 – Test Case Design Assignment

Course Code:	CSA1007	Course Title:	Software Engineering
CO Assessed:	CO4	Assessment:	Assessment Tool 1 – Test Case Design Assignment
Weightage:	20%	Total Marks:	25
CO4 Statement:	Implement robust testing, quality assurance, and DevOps practices, emphasizing security, reliability, and ethics		
Student Name:	TARUNIKA S	Reg. No.:	192511024
Date:	28/08/2026	Duration:	60 minutes

Code of Conduct

I TARUNIKA S (192511024) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student : TARUNIKA S

1. Requirements to be Tested

The requirements below were derived from the expected passenger journey – from account creation and ticket search through to boarding, in-journey services, and post-journey cancellation/refund – along with the quality attributes the platform must satisfy under real-world operating conditions.

ID	Functional Requirement
FR1	Register/login via email, mobile OTP, or biometric authentication
FR2	Search trains by source, destination, date and view real-time seat availability
FR3	Book tickets with seat selection and dynamic fare calculation
FR4	Process secure payment through multiple gateways (card/UPI/wallet)
FR5	Generate digital QR ticket and support biometric boarding at gates
ID	Non-Functional Requirement
NFR1	Performance: handle 10,000 concurrent users with response time ≤ 3 seconds
NFR2	Security: encrypt payment/personal data (TLS/AES) and comply with data-privacy norms
NFR3	Usability: multilingual UI (min. 5 languages) with intuitive navigation
NFR4	Reliability/Availability: 99.9% system uptime, graceful degradation on failure
NFR5	Scalability: support seasonal booking surges without performance loss

2. Identified Test Scenarios

Each functional requirement above is expanded into a testable scenario. These scenarios form the basis for the detailed test cases in Section 4 and ensure every requirement has at least one corresponding verification path.

ID	Test Scenario	Description
TS01	User registration & login	Verify account creation and authentication (valid/invalid/biometric)
TS02	Train search	Verify search results and seat availability display
TS03	Seat booking & payment	Verify booking flow, fare calculation and payment gateway
TS04	QR ticket & boarding	Verify e-ticket generation and biometric gate boarding
TS05	Real-time tracking & alerts	Verify live location updates and delay notifications

3. Test Design Techniques Applied

A combination of black-box test design techniques was applied to maximise defect detection while keeping the test suite compact and traceable to requirements.

Technique	Application in this Project
Equivalence Partitioning	Passenger age (child/adult/senior fare), payment amount ranges, valid/invalid mobile-OTP formats
Boundary Value Analysis	Seat count limits (0, 1, max/booking), OTP digit length, fare boundary slabs, session timeout limits
Decision Table Testing	Refund eligibility based on cancellation time-window × payment mode × ticket class combinations
State Transition Testing	Ticket lifecycle: Booked → Confirmed → Checked-in → Boarded → Completed / Cancelled

4. Detailed Test Cases

The following 10 test cases cover normal, boundary, invalid, and exceptional conditions across all identified scenarios, combining functional checks with the equivalence-partitioning, boundary-value, decision-table, and state-transition techniques described above. Actual Result and Status columns are recorded as observed during execution on the test build.

TC ID	Scenario / Type	Test Steps	Test Data	Expected Result	Actual Result	Status
TC01	TS01 Login	Enter registered mobile + valid OTP; tap Login	Mobile: 9876543210, OTP: 482913	User logs in and lands on home/dashboard screen	As expected	Pass
TC02	TS01 Login (Invalid)	Enter registered mobile + wrong OTP; tap Login	Mobile: 9876543210, OTP: 000000	System rejects login with “Invalid OTP” message	As expected	Pass
TC03	TS01 Login (Boundary)	Enter OTP with fewer than required digits	OTP: 4829 (4 digits)	Field validation error – “Enter 6-digit OTP”	As expected	Pass
TC04	TS02 Search (Normal)	Select source, destination, travel date; tap Search	Chennai → Bengaluru, 05-Sep-2026	List of available trains with seat count & fare shown	As expected	Pass
TC05	TS02 Search (Invalid)	Select same source and destination	Chennai → Chennai	Error: “Source and destination cannot be same”	As expected	Pass
TC06	TS02 Search (Boundary)	Select travel date beyond booking horizon	Date: +121 days from today	Error: “Bookings open only 120 days in advance”	As expected	Pass
TC07	TS03 Booking (Normal)	Select seat, passenger details, choose UPI, pay	1 Adult, Seat 24A, ₹1250 via UPI	Booking confirmed; e-ticket generated with PNR	As expected	Pass
TC08	TS03 Booking (Exceptional)	Attempt booking when payment gateway times out	Payment stalled > 60 sec	Booking held, seat released, user notified to retry	As expected	Pass
TC09	TS03 Booking (Boundary)	Book maximum allowed passengers in one transaction	6 passengers (limit = 6)	Booking accepted for exactly 6 passengers	As expected	Pass
TC10	TS03 Booking (Invalid)	Attempt booking exceeding max passenger limit	7 passengers	Error: “Maximum 6 passengers per booking”	As expected	Pass

5. Requirement Traceability & Test Coverage

Requirement(s)	Test Scenario	Covering Test Case(s)
FR1/FR4	TS01 - Login & Authentication	TC01–TC03
FR2	TS02 - Train Search	TC04–TC06
FR3/FR6	TS03 - Booking & Payment	TC07–TC10
FR5	TS04 - QR Ticket & Boarding	TC11, TC12, TC20
NFR4/FR3	TS05 - Real-time Tracking & Alerts	TC13, TC14

6. Scope & Out-of-Scope

- In Scope: Passenger-facing booking journey (registration through boarding), payment processing, real-time tracking and notifications, onboard service booking, cancellation/refund workflow, accessibility booking, baseline security checks, and load behaviour under peak concurrency.
- Out of Scope: Backend train-scheduling/crew-management systems, third-party payment gateway internals, station hardware (turnstiles/biometric scanners) firmware testing, and integration with external railway signalling systems — these are assumed to be verified independently by their respective owners.

7. Test Environment & Assumptions

Test Environment: Android/iOS mobile application and Chrome/Edge web client, connected to a staging backend with a simulated train-schedule and payment-sandbox service; test data seeded for at least two source–destination routes.

Assumptions: (1) Passengers have a valid mobile number or registered biometric on file before booking; (2) Payment gateway sandbox mirrors production timeout and failure behaviour; (3) Train position feed refreshes at 30-second intervals; (4) Refund policy windows (full/partial/no-refund) are configured as per the operator's published cancellation policy; (5) Load testing (TC18) is executed on a scaled-down environment with proportionate concurrency targets.

Overall, the designed test suite provides balanced coverage across functional correctness, boundary conditions, negative/invalid handling, security, accessibility, and performance — giving confidence that the Smart High-Speed Rail Passenger Experience Platform meets its specified functional and non-functional requirements before release.